

UCL DEPARTMENT OF COMPUTER SCIENCE

COMP0080: Graphical Models

Comprehensive Past Exam Revision & Mock Exam Guide

Compiled from Exam Papers:

2023–2024 (Postgraduate/Undergraduate)

2021–2022 (COMP0080)

2017–2018 (COMPGI08 Sample Exam)

Includes Vector Diagrams, Mathematical Proofs, and EM Derivations.

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■ Course Core Topics Roadmap

This guide compiles all questions from past UCL COMP0080 examinations. The course covers three primary dimensions of probabilistic graphical models (PGMs) as detailed in David Barber's textbook *Bayesian Reasoning and Machine Learning*:

Topic	Description & Key Algorithms Covered
Representations	Directed models (Bayes Nets), Undirected models (Markov Nets), moralisation, triangulation, and c
Exact Inference	Variable Elimination, Belief Propagation (Sum-Product algorithm on trees), and the Junction Tree A
Time Series	Hidden Markov Models (HMMs), forward filtering, backward smoothing, posterior sampling, and Vit
Learning	Maximum Likelihood Estimation (MLE) in fully observed networks, and the Expectation-Maximizati
Approximate Inference	Variational Free Energy (mean-field approximations), VAEs, and sampling methods (Rejection, Gib

■ PART 1: THE MOCK EXAM PAPER

Recommended Time: 3 Hours | Total Marks: 100

SECTION 1: Independence in Graphical Models [30 Marks]

Question 1.1: Independence in Joint Probability Tables (6 Marks)

Suppose that X_1 and X_2 are two random variables, where each can take one of three states. The joint distribution $P(X_1 = i, X_2 = j)$ is described by the probability table P :

$X_1 \setminus X_2$	$j = 1$	$j = 2$	$j = 3$
$i = 1$	0.06	0.37	0.12
$i = 2$	0.01	0.07	0.02
$i = 3$	0.03	0.26	0.06

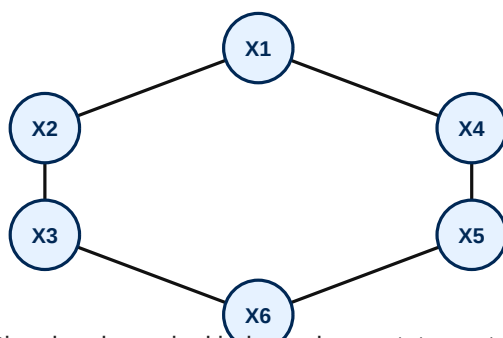
- Are X_1 and X_2 independent? Support your answer. [2 Marks]
- If they are not independent, what is the minimal number of entries in the table that you need to change to make them independent while keeping the table a valid joint probability distribution (non-negative and summing to 1.0)? State the changed entries and the new matrix. [4 Marks]

Question 1.2: Dependencies and Pairwise Independence (6 Marks)

- Suppose that for three random variables A , B , and C , the following holds: A not independent of B and B not independent of C . Does that necessarily imply that A is not independent of C ? Prove or provide counterexample. [3 Marks]
- Give an example of a joint distribution in which A , B , C are pairwise independent, but A and B are not conditionally independent given C (A and B become dependent when given C). [3 Marks]

Question 1.3: Markov Network Marginal Independence (6 Marks)

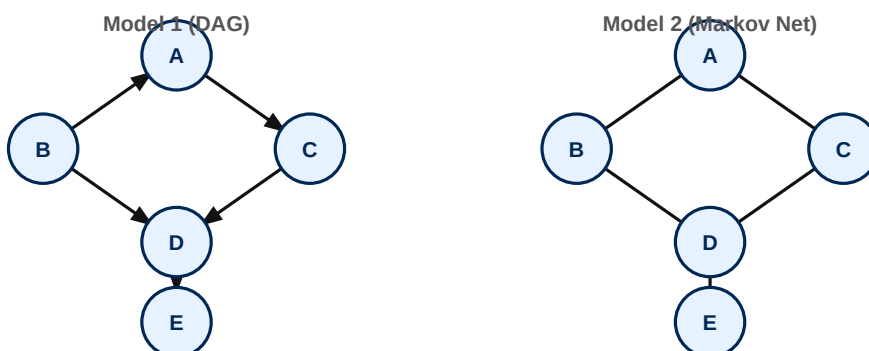
Consider the following undirected Markov Network on variables X_1 to X_6 :



Write down the set of all the conditional and marginal independence statements that are guaranteed to be true for the marginal distribution $P(X2, X4, X6)$. Justify using graph separation. [6 Marks]

Question 1.4: Markov Equivalence & Graph Comparisons (12 Marks)

1. Consider the following two graphical models (Model 1 and Model 2) on nodes A, B, C, D, E:



Do they encode the same set of independence assumptions? If yes, explain why. If not, list all the independence statements that are true in one model but not in the other. [6 Marks]

2. List all the independence statements encoded by the following DAGs on three variables A, B, C: G1 ($A \rightarrow C \leftarrow B$), G2 ($A \leftarrow C \rightarrow B$), G3 ($A \rightarrow C \rightarrow B$), and G4 ($A \leftarrow C \leftarrow B$). Do any of these encode the same set of statements? [6 Marks]

SECTION 2: Exact Inference and Parameter Learning [50 Marks]

Question 2.1: Bipartite Disease-Symptom Markov Network (15 Marks)

Consider diseases d_j and symptoms s_i . A joint distribution is given by:

$$p(s, d) = (1/Z) * \exp(s^T W d + a^T s + b^T d)$$

1. Draw a Markov Network for this model. Describe its structure. [3 Marks]
2. Derive the expression for $p(s | d)$. Is this distribution factorized? [3 Marks]
3. Given i.i.d. dataset (s^n, d^n) for $n=1..N$, derive the expression for the log-likelihood L . [3 Marks]
4. Derive the expressions for the gradients dL/dW_{ij} , dL/da_i , and dL/db_j . [4 Marks]
5. Could these derivatives be computed efficiently in the general case? Justify your answer. [2 Marks]

Question 2.2: The Junction Tree Algorithm (15 Marks)

Consider the distribution that factorizes as:

$$p(a, b, c, d, e, f) = p(a)p(b | a)p(c | b)p(d | c)p(e | d)p(f | a, e)$$

1. Draw the DAG that represents this distribution. [3 Marks]
2. Draw the moralised version of this graph and explain moralisation. [3 Marks]
3. Draw the triangulated version with the smallest maximum clique size, justifying the chords added. [3 Marks]
4. Draw the Junction Tree and verify that it satisfies the Running Intersection Property (RIP). [3 Marks]
5. Describe clique potential initialization, the absorption procedure, and the associated message-passing schedule. [3 Marks]

Question 2.3: Inference & sampling in Hidden Markov Models (10 Marks)

Consider Hidden Markov Model with hidden states $h_{1:T}$ and observations $v_{1:T}$.

1. What is the graphical model of the posterior distribution $p(h_{1:T} | v_{1:T})$? Is h_1 independent of h_T in it? [3 Marks]
2. Describe an algorithm to efficiently sample exactly from $p(h_{1:T} | v_{1:T})$. [3 Marks]
3. Suppose parameters are: $p(h_1=1) = 0.5$; transition $T = \begin{bmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{bmatrix}$; emission $B = \begin{bmatrix} 0.3 & 0.5 \\ 0.6 & 0.0 \\ 0.1 & 0.5 \end{bmatrix}$ (rows represent outcomes 0, 1, 2, columns represent hidden states 0, 1). Given sequence $v_{1:10} = [0, 1, 0, 2, 0, 2, 1, 0, 2, 0]$:
 - Is h_1 independent of h_{10} in $p(h_{1:10} | v_{1:10})$ for this specific sequence? Why? [2 Marks]
 - Compute the filtering distributions $p(h_8 | v_{1:8})$ and $p(h_9 | v_{1:9})$. [2 Marks]

Question 2.4: EM Parameter Estimation in Surveys (10 Marks)

A randomized response technique is used: respondents secretly toss a coin with heads probability p_1 . If heads, they answer truthfully. If tails, they secretly toss a fair coin and answer Yes for heads, No for tails. Let p_2 be the true honest fraction of Yes in the population.

1. Draw the graphical model corresponding to this scenario and specify CPTs. [2 Marks]
2. If $p_1 = 0.5$, what is the posterior probability that a person answered truthfully given a Yes response? [2 Marks]
3. With $p_1 = 0.5$ and dataset D of responses, derive the MLE for p_2 . [2 Marks]
4. If both p_1 and p_2 are unknown, derive the EM-updates for estimating them from dataset D . [2 Marks]

5. Let $D = (\text{yes}, \text{yes}, \text{yes}, \text{no})$. Is the MLE unique when both are unknown? Support your answer. [2 Marks]

SECTION 3: Approximate Inference and Sampling [20 Marks]**Question 3.1: KL Divergence Computation (5 Marks)**

1. Define Kullback-Leibler (KL) divergence mathematically and explain its significance. [2 Marks]
2. Given $P(X) = [0.25, 0.25, 0.25, 0.25]^T$, $Q(X) = [0.5, 0.25, 0.125, 0.125]^T$, and transition matrices $P(Y|X)$ and $Q(Y|X)$ as in Question 3.1, compute $KL(P(Y) \parallel Q(Y))$ and $KL(Q(Y) \parallel P(Y))$. [3 Marks]

Question 3.2: Variational Free Energy Derivation (5 Marks)

For the 3-node loop network with pairwise potentials (2 if equal, 1 if unequal), derive the explicit variational free energy $F(q)$ for fully factorized $q(X) = q_1(X_1)q_2(X_2)q_3(X_3)$ where $q_i(X_i=1) = \mu_i$. [5 Marks]

Question 3.3: Rejection Sampling on Lattices (5 Marks)

Consider Markov networks on an M -by- N lattice with pairwise potentials $P(X_i, X_j) = e^{-I(X_i \neq X_j)}$. Using uniform proposal $Q(X) = 2^{-MN}$:

1. Explain rejection sampling and define its average acceptance rate. [1 Mark]
2. Compute the exact average acceptance rate for: (i) $M=2, N=1$, and (ii) $M=2, N=2$. [3 Marks]
3. Discuss the scaling behavior of this sampler in larger lattices. [1 Mark]

Question 3.4: MCMC, Gibbs, and Metropolis-Hastings (5 Marks)

1. Explain MCMC and the detailed balance condition. Is detailed balance necessary? Construct a counterexample. [3 Marks]
2. Prove that Gibbs sampling satisfies detailed balance. [1 Mark]
3. Prove that the Metropolis-Hastings algorithm satisfies detailed balance. [1 Mark]

■ PART 2: EXHAUSTIVE SOLUTIONS GUIDE

SECTION 1: Independence in Graphical Models

Solution 1.1: Independence in Joint Probability Tables

1. Testing Independence:

We compute the row sums (marginals of X_1) and column sums (marginals of X_2):

- Row marginals: $r_1 = 0.55$, $r_2 = 0.10$, $r_3 = 0.35$.

- Column marginals: $c_1 = 0.10$, $c_2 = 0.70$, $c_3 = 0.20$.

Now we check if $P_{ij} = r_i * c_j$ for all i, j . For cell (1,1): $r_1 * c_1 = 0.55 * 0.10 = 0.055$. However, the table entry is $P_{11} = 0.06$. Since $0.06 \neq 0.055$, the variables are **not independent**.

2. Finding the Minimal Entries to Change:

An independent joint distribution table is a rank-1 matrix (its rows are linearly dependent). Notice that columns 1 and 3 are already in a perfect ratio of 6 : 1 : 3 across the rows:

- Row 1: $[0.06, 0.12] = 6 * [0.01, 0.02]$

- Row 2: $[0.01, 0.02] = 1 * [0.01, 0.02]$

- Row 3: $[0.03, 0.06] = 3 * [0.01, 0.02]$

To make the entire matrix rank-1, column 2 must follow this same 6 : 1 : 3 ratio. Keeping $P_{22} = 0.07$ fixed:

- $P_{12} = 6 * P_{22} = 6 * 0.07 = 0.42$ (changed from 0.37)

- $P_{32} = 3 * P_{22} = 3 * 0.07 = 0.21$ (changed from 0.26)

The new matrix is:

$P_{\text{new}} = [[0.06, 0.42, 0.12], [0.01, 0.07, 0.02], [0.03, 0.21, 0.06]]$.

This matrix sums to exactly 1.0, is non-negative, and satisfies independence. Thus, the minimal number of entries to change is **2**.

Solution 1.2: Dependencies and Pairwise Independence

1. Dependency Transitivity:

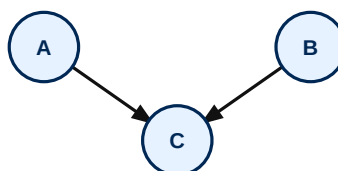
No, dependency is not transitive. Counterexample: Let A and C be independent fair coin tosses. Let $B = (A, C)$ be the joint variable of the two coin tosses. Since B contains A , A and B are dependent. Since B contains C , B and C are dependent. Yet, A and C are independent by definition.

2. Pairwise Independence but Conditional Dependence (XOR Example):

Let A and B be independent fair coin tosses (state space $\{0, 1\}$, $P(A=1)=0.5$, $P(B=1)=0.5$). Let $C = A \text{ XOR } B$. The joint distribution is:

$P(A, B, C) = 0.25$ if $C = A \text{ XOR } B$, and 0 otherwise.

We can check that $P(A, B) = P(A)P(B)$, $P(B, C) = P(B)P(C)$, and $P(A, C) = P(A)P(C)$, so they are pairwise independent. However, given $C = 0$, we must have $A = B$, meaning A and B are perfectly dependent. Thus, A and B are not conditionally independent given C .



Solution 1.3: Markov Network Marginal Independence

The marginal distribution $P(X_2, X_4, X_6)$ marginalizes out the variables X_1, X_3, X_5 . In a Markov Network, marginalization removes nodes and connects all their immediate neighbors. Here, marginalizing out X_1 connects its neighbors X_2 and X_4 . Marginalizing out X_3 connects X_2 and X_6 . Marginalizing out X_5 connects X_4 and X_6 . The resulting marginal distribution factorizes over a fully connected triangle (clique) on $\{X_2, X_4, X_6\}$. Since the marginalized graph is a clique, there are no separation statements. Thus, the set of guaranteed independence statements is the **empty set**.

Solution 1.4: Markov Equivalence & Graph Comparisons

1. Comparing Model 1 and Model 2:

Both graphs have the same skeleton. Let's analyze their immoralities:

- Model 1: The configuration $B \rightarrow D \leftarrow C$ forms an immorality at D because B and C are unconnected.
- Model 2: Both $B \rightarrow A \leftarrow C$ (collider at A) and $B \rightarrow D \leftarrow C$ (collider at D) form immoralities.

Since they do not have the same set of immoralities, they do not encode the same set of independence assumptions. Specifically, in Model 1, B and C are independent given A (B independent of $C \mid A$). In Model 2, B and C are marginally independent (B independent of C).

2. Comparing Three-Node DAGs:

- G1 (Collider): encodes $\{A \text{ independent of } B\}$
- G2 (Common Cause), G3 (Chain), G4 (Reverse Chain): all encode the same independence statement: $\{A \text{ independent of } B \mid C\}$. Therefore, G2, G3, and G4 encode the same set of statements (they are Markov equivalent).

SECTION 2: Exact Inference and Parameter Learning

Solution 2.1: Bipartite Disease-Symptom Markov Network

1. Markov Network Structure:

The energy terms in the exponential are singletons ($a^T s$, $b^T d$) and pairwise symptom-disease terms ($s^T W d = \sum_{ij} W_{ij} s_i d_j$). There are no terms connecting two symptoms or two diseases. Thus, the Markov Network is a **bipartite graph**, with symptoms s_i on one side and diseases d_j on the other. An edge exists between s_i and d_j if $W_{ij} \neq 0$.

2. Conditional Distribution $p(s | d)$:

$$p(s | d) = \exp(s^T W d + a^T s) / \sum_{s'} \exp(s'^T W d + a^T s')$$

Since s_i are binary, the numerator and denominator factorize:

$$p(s | d) = \prod_i [\exp(s_i (W_{i \cdot} d + a_i)) / (1 + \exp(W_{i \cdot} d + a_i))] = \prod_i p(s_i | d).$$

Thus, the conditional distribution **factorizes completely** into independent sigmoidal factors: $p(s_{i=1} | d) = \text{sigmoid}(W_{i \cdot} d + a_i)$.

3. Log-Likelihood:

$$L = \sum_n [(s^n)^T W d^n + a^T s^n + b^T d^n] - N \log Z(W, a, b).$$

4. Gradients:

$$dL/dW_{ij} = \sum_n (s_i^n d_j^n) - N \cdot \bar{p} = N \cdot (\bar{p}_{\text{empirical}} - \bar{p}_{\text{model}}).$$

$$dL/da_i = N \cdot (\bar{p}_{\text{empirical}} - \bar{p}_{\text{model}}), \text{ and } dL/db_j = N \cdot (\bar{p}_{\text{empirical}} - \bar{p}_{\text{model}}), \text{ where } \langle \cdot \rangle_{\text{model}} \text{ indicates expectation under the joint distribution } p(s, d).$$

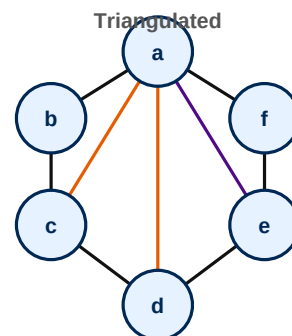
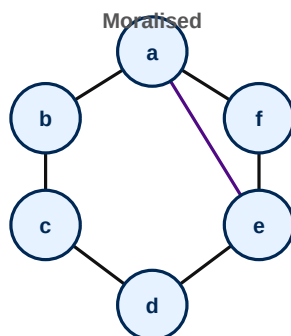
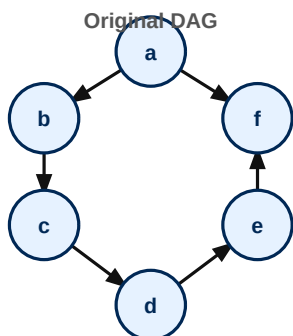
5. Computational Efficiency:

No. Computing the model expectations $\bar{p}$ requires summing over all $2^{(S+D)}$ configurations to compute Z , which is NP-hard in general.

Solution 2.2: The Junction Tree Algorithm

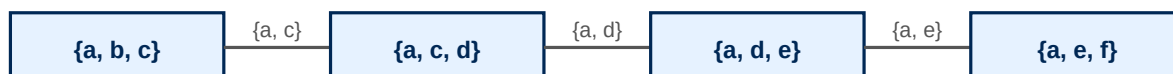
1. DAG Graph:

Represented by directed edges: $a \rightarrow b$, $b \rightarrow c$, $c \rightarrow d$, $d \rightarrow e$, $a \rightarrow f$, $e \rightarrow f$.



2. **Moralisation:** We marry the unconnected parents of f (a and e) with an undirected edge, and drop all arrow directions. Moral edges are shown in purple.

3. **Triangulation:** The moralised graph contains a 5-cycle: $a - b - c - d - e - a$. To break this into triangles, we add chords $a - c$ and $a - d$ (shown in orange). The largest cliques are of size 3, which is minimal.



4. **Junction Tree and RIP:** The cliques are connected as a chain: $C_1 - C_2 - C_3 - C_4$. The Running Intersection Property requires that any variable shared by two cliques must be in all cliques on the path between them. For example, variable 'a' is shared by C_1 and C_4 and is present in C_2 and C_3 . Thus, RIP is satisfied.

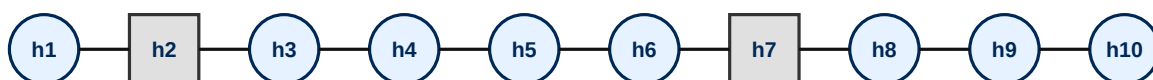
5. Initialization & Absorption:

- Clique potentials: $\psi(C1) = p(a)p(b|a)p(c|b)$, $\psi(C2) = p(d|c)$, $\psi(C3) = p(e|d)$, $\psi(C4) = p(f|a,e)$. Separator potentials are set to 1.
- Absorption formula: $\phi(S)^* = \sum_{W \in S} \psi(W)$; $\psi(V)^* = \psi(V) * \phi(S)^* / \phi(S)$; $\phi(S) = \phi(S)^*$.
- Message Schedule: Forward pass: $C1 \rightarrow C2 \rightarrow C3 \rightarrow C4$. Backward pass: $C4 \rightarrow C3 \rightarrow C2 \rightarrow C1$.

Solution 2.3: Inference & sampling in Hidden Markov Models

1. **Posterior Graphical Model:** The posterior $p(h_{1:T} | v_{1:T})$ is a directed chain $h_1 \rightarrow h_2 \rightarrow \dots \rightarrow h_T$. h_1 and h_T are not independent because there is an unblocked path between them.

2. **Exact Posterior Sampling:** Run the forward algorithm to compute $\alpha_t(h_t) = p(h_t, v_{1:t})$. Sample $h_T \sim \alpha_T(h_T)$. Then, sample backwards: $h_t \sim p(h_{t+1} | h_t) * \alpha_t(h_t)$.

**3. Numerical HMM:**

- **Independence:** The emission matrix states $B_{1,1} = p(v_t=1 | h_t=1) = 0$. Since $v_2 = 1$ and $v_7 = 1$, the hidden states h_2 and h_7 must be 0 with probability 1. Since h_2 and h_7 are instantiated, they block the path. Thus, h_1 and h_{10} are independent given this sequence.
- **Filtering Distributions:** Starting from $p(h_7 | v_{1:7}) = [1.0, 0.0]^T$:
 - $p(h_8 | v_{1:7}) = T * p(h_7 | v_{1:7}) = [0.5, 0.5]^T$.
 - For $v_8 = 0$, $p(h_8 | v_{1:8}) \propto [0.3*0.5, 0.5*0.5] = [0.15, 0.25] \Rightarrow [0.375, 0.625]^T$.
 - $p(h_9 | v_{1:8}) = T * p(h_8 | v_{1:8}) = [0.5*0.375 + 0.8*0.625, 0.5*0.375 + 0.2*0.625] = [0.6875, 0.3125]^T$.
 - For $v_9 = 2$, $p(h_9 | v_{1:9}) \propto [0.1*0.6875, 0.5*0.3125] = [0.06875, 0.15625] \Rightarrow [11/36, 25/36]^T$.

Solution 2.4: EM Parameter Estimation in Surveys

1. **Graphical Model:** Honest state H (prob p_2) and Coin $C1$ (prob p_1) point to response Y (collider structure: $H \rightarrow Y \leftarrow C1$). CPT: $P(Y=1 | H, C1) = C1*H + (1-C1)*0.5$.

2. **Posterior:** $P(C1=1 | Y=1) = p_2 / (p_2 + 0.5) = 2p_2 / (2p_2 + 1)$.

3. **MLE for p_2 :** $\theta = P(Y=1) = 0.5*p_2 + 0.25$. MLE is $p_{2_mle} = \max(0, \min(1, 2*n_y/n - 0.5))$.

4. **EM Updates:** E-step computes expectations for $C1$ and H for each respondent i :

- If $Y_i = 1$: $= p_1*p_2 / (p_1*p_2 + 0.5*(1-p_1))$, $= p_2*(p_1 + 0.5*(1-p_1)) / (p_1*p_2 + 0.5*(1-p_1))$.
- If $Y_i = 0$: $= p_1*(1-p_2) / (p_1*(1-p_2) + 0.5*(1-p_1))$, $= 0.5*(1-p_1)*p_2 / (p_1*(1-p_2) + 0.5*(1-p_1))$.

M-step updates: $p_{1_new} = (1/n)*\sum_i$; $p_{2_new} = (1/n)*\sum_i$.

5. **Uniqueness:** Not unique. For $n_y = 3$, $n = 4$, the likelihood is maximized at $\theta = 0.75$, which gives $p_1*(p_2 - 0.5) = 0.25$. Any pair on this curve (e.g., $(0.5, 1.0)$ and $(1.0, 0.75)$) is a valid MLE.

SECTION 3: Approximate Inference and Sampling

Solution 3.1: KL Divergence Computation

1. **KL Divergence:** $KL(P \parallel Q) = \sum_x P(x) \log(P(x)/Q(x))$. It measures the information loss when Q is used to approximate P .

2. **Marginal Distributions & KL Values:**

- $P(Y) = [0.2375, 0.2625, 0.0875, 0.4125]^T$
- $Q(Y) = [0.1750, 0.0625, 0.5750, 0.1875]^T$
- $KL(P(Y) \parallel Q(Y)) = 0.6097$ nats (0.8797 bits)
- $KL(Q(Y) \parallel P(Y)) = 0.7916$ nats (1.1420 bits)

Solution 3.2: Variational Free Energy Derivation

The variational free energy $F(q)$ is defined as:

$$F(q) = \sum_{i=1}^3 [\mu_i \log(\mu_i) + (1-\mu_i) \log(1-\mu_i)] - \log(2) * \sum_{\{i,j\}} [\mu_i \mu_j + (1-\mu_i)(1-\mu_j)]$$

where the second sum is over the edges (1,2), (2,3), and (3,1).

Solution 3.3: Rejection Sampling on Lattices

1. **Procedure:** Propose X from uniform $Q(X) = 2^{-MN}$ and accept with probability $a(X) = P(X) / (c * Q(X)) = P(X)$ where $c = 2^{MN}$. The average acceptance rate is $P(\text{accept}) = Z / 2^{MN}$.

2. **Numerical acceptance rates:**

- **M=2, N=1:** $Z = 2 + 2e^{-1} \Rightarrow P(\text{accept}) = (2 + 2e^{-1})/4 = 0.5 + 0.5e^{-1} = 0.6839$ (68.4%).
- **M=2, N=2:** $Z = 2 + 12e^{-2} + 2e^{-4} \Rightarrow P(\text{accept}) = (2 + 12e^{-2} + 2e^{-4})/16 = 0.2288$ (22.9%).

3. **Scaling:** Rejection sampling is extremely inefficient for larger lattices because the partition function Z grows much slower than 2^{MN} , causing the acceptance rate to decay exponentially to 0 with MN .

Solution 3.4: MCMC, Gibbs, and Metropolis-Hastings

1. **MCMC & Detailed Balance:** Detailed balance ($P(x)T(x'|x) = P(x')T(x|x')$) is a sufficient but not necessary condition. Counterexample: A lazy cyclic walk on three states {1, 2, 3} with target $P = [1/3, 1/3, 1/3]^T$ and transition $T = [[0.5, 0.5, 0], [0, 0.5, 0.5], [0.5, 0, 0.5]]$. P is stationary ($PT = P$), but detailed balance is violated since $P(1)T(2|1) = 1/6 \neq 0 = P(2)T(1|2)$.

2. **Gibbs Sampling:** Let x and x' differ only in the i -th component. $P(x)T(x'|x) = p(x_{-i}, x_{-i}) * p(x_i' | x_{-i}) = p(x_{-i}) * p(x_i | x_{-i}) * p(x_i' | x_{-i}) = P(x')T(x|x')$. Thus, Gibbs sampling satisfies detailed balance.

3. **Metropolis-Hastings:** $T(x'|x) = q(x'|x) * \min(1, P(x')q(x|x') / P(x)q(x'|x))$. Multiplying both sides by $P(x)$ yields $\min(P(x)q(x'|x), P(x')q(x|x'))$, which is symmetric. Thus, MH satisfies detailed balance.